Annex B. Production environment variables

Version 1.0 · October 28, 2026. Variables that `docker-compose.prod.yml` takes from `infra/.env.prod` and the service that uses them, read from the file on `main` of `energy-monitor-infra`. Only names and purpose are documented: no value appears here. Secrets are generated on the server (`openssl rand -hex 24`), live only in `.env.prod` (permissions 600) and go to no repository.

Variables by purpose

Variable	Type	Services	Purpose
`EXPO_ACCESS_TOKEN`	Secret	`notification`	Expo token for mobile push
`GOOGLE_CLIENT_SECRET`	Secret	`security`	Google OAuth client secret
`INTERNAL_API_TOKEN`	Secret	`security`, `device`, `measurement`, `home`, `notification`, `alert`, `recommendation`	`X-Internal-Token` header of the internal calls between services
`MAIL_PASSWORD`	Secret	`notification`	Application password of the email account
`MQTT_DEVICE_SERVICE_PASS`	Secret	`device`	MQTT password of the device service
`MQTT_MEASUREMENT_SERVICE_PASS`	Secret	`device`, `measurement`	MQTT password of the measurement service
`MYSQL_PASSWORD`	Secret	`mysql`, `security`, `device`, `measurement`, `home`, `notification`, `alert`, `recommendation`	Password of the MySQL `energy_app` user
`MYSQL_ROOT_PASSWORD`	Secret	`mysql`	MySQL `root` password
`NOTIFICATION_VAPID_PRIVATE_KEY`	Secret	`notification`	Web Push VAPID private key (single line)
`RABBITMQ_PASSWORD`	Secret	`rabbitmq`, `security`, `device`, `measurement`, `home`, `notification`, `alert`, `recommendation`	RabbitMQ password, the same in every service
`SECURITY_RESET_PEPPER`	Secret	`security`	HMAC pepper of the codes (verification, recovery and deletion); changing it invalidates the codes already sent
`ALERT_LIMIT_ZONE`	Configuration	`alert`	Time zone of the daily and monthly limits
`BACKEND_GID`	Configuration	`security`	Group the security service runs as
`BACKEND_UID`	Configuration	`security`	User the security service runs as, owner of the keys
`DEVICE_MQTT_PUBLIC_HOST`	Configuration	`device`	Broker domain the web app writes to the meter (`mqtt.energy-monitor.tech`)
`DEVICE_MQTT_PUBLIC_PORT`	Configuration	`device`	Broker TLS port (8883)
`GOOGLE_CLIENT_ID`	Configuration	`security`	Google OAuth client (public)
`MAIL_FROM`	Configuration	`notification`	Sender of the emails

Variable	Type	Services	Purpose
`MAIL_HOST`	Configuration	`notification`	SMTP server
`MAIL_PORT`	Configuration	`notification`	SMTP port
`MAIL_USERNAME`	Configuration	`notification`	Sending email account
`MQTT_DEVICE_SERVICE_USER`	Configuration	`device`	MQTT user of the device service
`MQTT_MEASUREMENT_SERVICE_USER`	Configuration	`device`, `measurement`	MQTT user of the measurement service
`NOTIFICATION_VAPID_PUBLIC_KEY`	Configuration	`notification`	Web Push VAPID public key
`NOTIFICATION_WEB_URL`	Configuration	`notification`	Public address for the links in the emails
`RABBITMQ_USERNAME`	Configuration	`rabbitmq`, `security`, `device`, `measurement`, `home`, `notification`, `alert`, `recommendation`	RabbitMQ user, the same in every service
`RECOMMENDATION_INITIAL_DELAY`	Configuration	`recommendation`	Wait before the first evaluation
`RECOMMENDATION_ZONE`	Configuration	`recommendation`	Time zone of the recommendations
`SECURITY_CORS_ALLOWED_ORIGINS`	Configuration	`gateway`	Origins allowed by the gateway
`VITE_GOOGLE_CLIENT_ID`	Configuration	`frontend`	Google client of the web button; baked in when the image is built
`WEB_PORT`	Configuration	`frontend`	Local port Nginx points to (8081)
`JWT_KEYS_DIR`	Path	`security`, `device`, `measurement`, `home`, `notification`, `alert`, `recommendation`, `gateway`	Folder of `private.pem` (security only) and `public.pem` (the others)
`MQTT_CERT_DIR`	Path	`mosquitto`	Folder with the broker's Let's Encrypt certificate
`ACCOUNT_DELETED_CONSUMER_ENABLED`	Switch	`home`	Cleans the data of a deleted account; `true` by default
`ACTIVITY_CONSUMER_ENABLED`	Switch	`notification`	Consumes home and device activity; `true` by default
`ALERT_CONSUMERS_ENABLED`	Switch	`alert`	Consumes readings and device events to create alerts; `true` by default
`ALERT_EVENTS_CONSUMER_ENABLED`	Switch	`notification`	Consumes `alert.raised` and `recommendation.created`; `true` by default
`ALERT_SERVICE_OUTBOX_RELAY_ENABLED`	Switch	`alert`	Publishes `alert.raised` from the *outbox*; `true` by default
`HOME_OUTBOX_RELAY_ENABLED`	Switch	`home`	Publishes the home *outbox* events; `true` by default
`RECOMMENDATION_OUTBOX_RELAY_ENABLED`	Switch	`recommendation`	Publishes `recommendation.created` from the *outbox*; `true` by default
`RECOMMENDATION_SCHEDULER_ENABLED`	Switch	`recommendation`	Hourly evaluation of the rules; `true` by default
`SECURITY_ACCOUNT_DELETION_CODE_MAIL_ENABLED`	Switch	`security`	Email with the deletion code of Google accounts; turn on only with notifications deployed (off, asking for the code answers 503)

Variable	Type	Services	Purpose
`SECURITY_EVENTS_CONSUMER_ENABLED`	Switch	`notification`	Consumes the account events and the mail; `true` by default

Clone paths

Each service is built from a clone next to `infra`. The variables `ALERT_CONTEXT`, `DEVICE_CONTEXT`, `FRONTEND_CONTEXT`, `GATEWAY_CONTEXT`, `HOME_CONTEXT`, `MEASUREMENT_CONTEXT`, `NOTIFICATION_CONTEXT`, `RECOMMENDATION_CONTEXT`, `SECURITY_CONTEXT` change that path; by default they point to `../backend/<repository>` and `../frontend/energy-monitor-web`, so they are only set if the clones are elsewhere.

Alert service properties (not `.env.prod` variables)

Property	Default	What it controls
`alert.anomaly.window`	`P7D`	The days of history from which the usual power of a device is computed for the unusual consumption alert
`alert.anomaly.baseline-ttl`	`PT10M`	How long that usual power is reused before asking the measurement service again

The thresholds of the alert (2.5 times, 200 W, 24 hours of history and a 5 W minimum average) are constants of the service, not properties.

Notes

- `.env.prod.example` keeps, by inheritance, the `BACKEND_CONTEXT` line of the retired backend and does not list `ALERT_CONTEXT` or `RECOMMENDATION_CONTEXT`: it has no effect, because the compose file defines their default, but the template should be aligned.
- `VITE_*` variables are baked in when the frontend image is built: changing them requires rebuilding `frontend`.
- A switch that was set to `false` because of the migration must be deleted from `.env.prod` (they are `true` by default).

Spanish version: ../ES/anexo-b-variables-de-entorno.md